



IE 1044 – Digital Electronics

**Assignment : Design a Mini-Processor for BCD Arithmetic
Operations**

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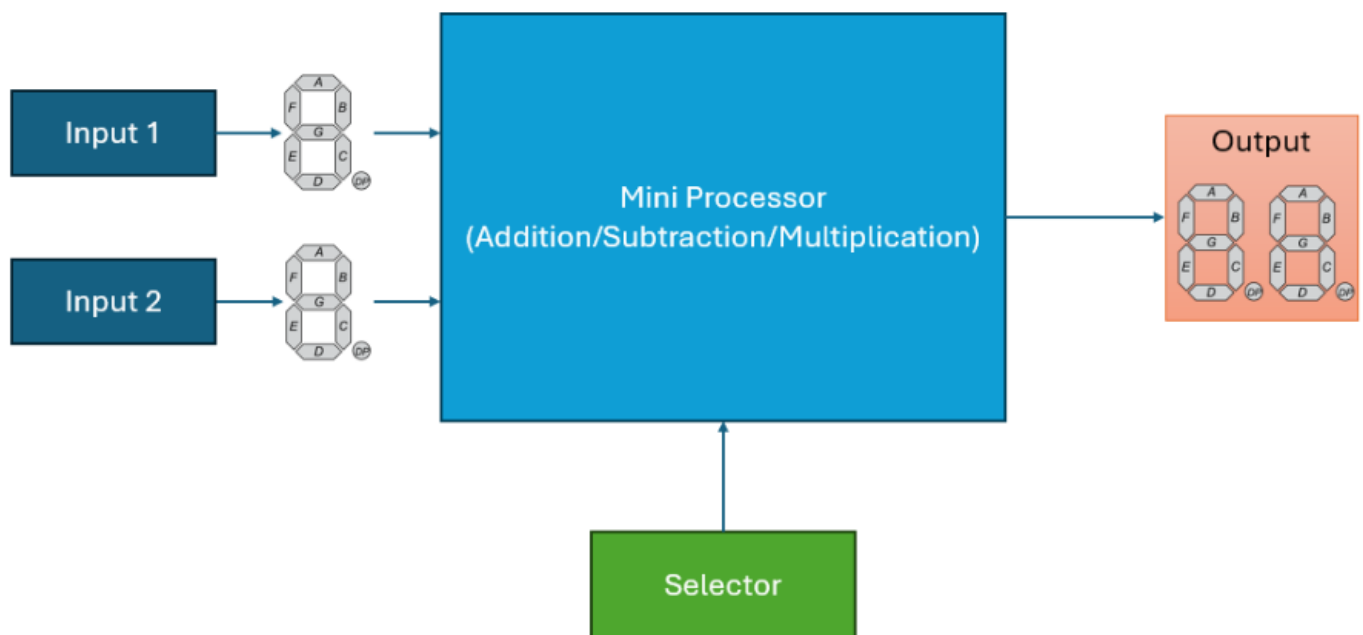
Year 01, Semester 02

Introduction

Binary Coded Decimal (BCD) is a form of number representation where each digit of a decimal number is represented by its own binary sequence. This approach is especially useful in digital systems where precise decimal representation is crucial, such as in financial and commercial applications.

This assignment involved designing a Mini-Processor capable of executing three arithmetic operations—Addition, Subtraction, and Multiplication—on 1-digit BCD inputs. The processor takes two BCD inputs, a 2-bit control signal to determine the operation, and outputs a 2-digit BCD result.

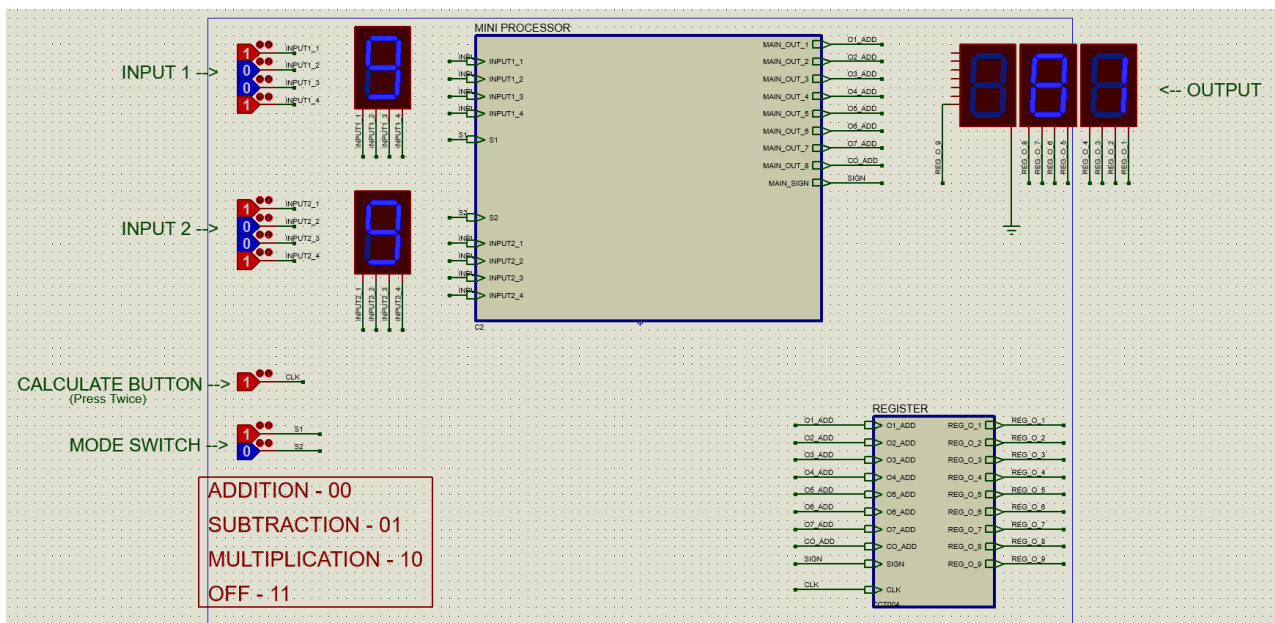
The objective of this project was to help understand the design of BCD arithmetic circuits, control logic implementation, and simulation using Proteus.



System Architecture and Component Design

The system consists of the following primary modules :

- BCD Adder with correction logic
- BCD Subtractor using two's complement and correction
- BCD Multiplier with Binary to BCD Converter
- Multiplexer for operation control
- Registers for Output Storage

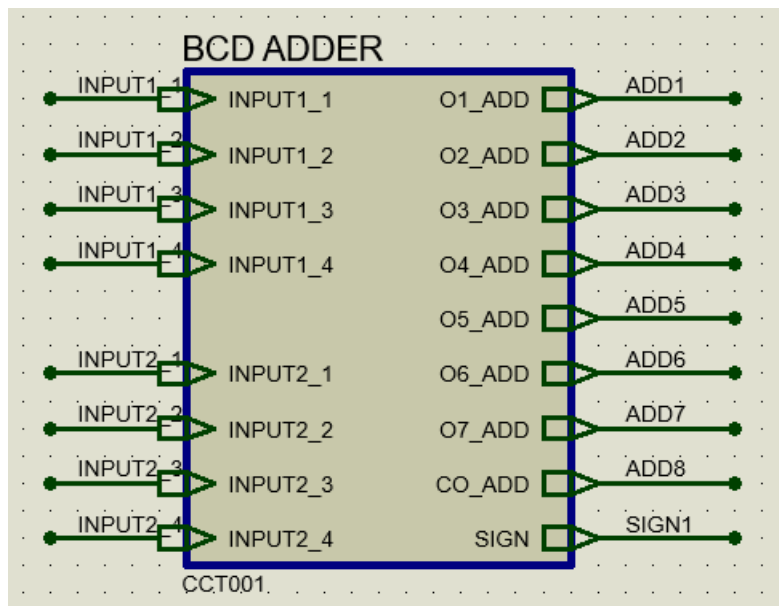


(Main Interface)

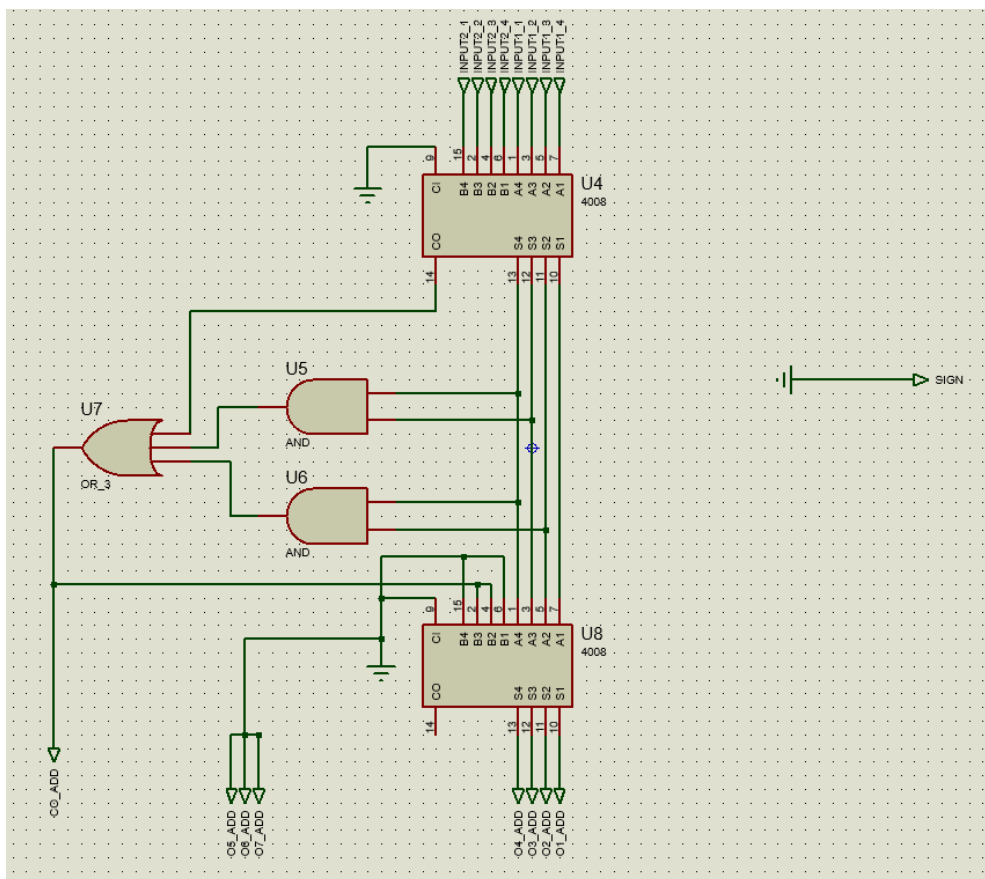
Instructions are given in the main interface to maintain ease for the user to use the calculator.

Mini Processor contains BCD Adder, Subtractor, Multiplier and Multiplexer to control operation.

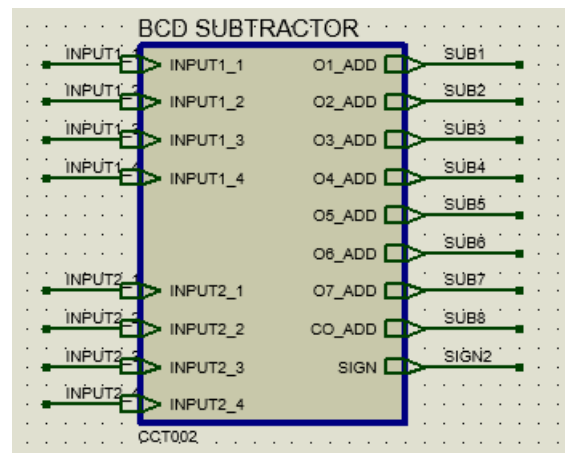
BCD Adder with Correction Logic



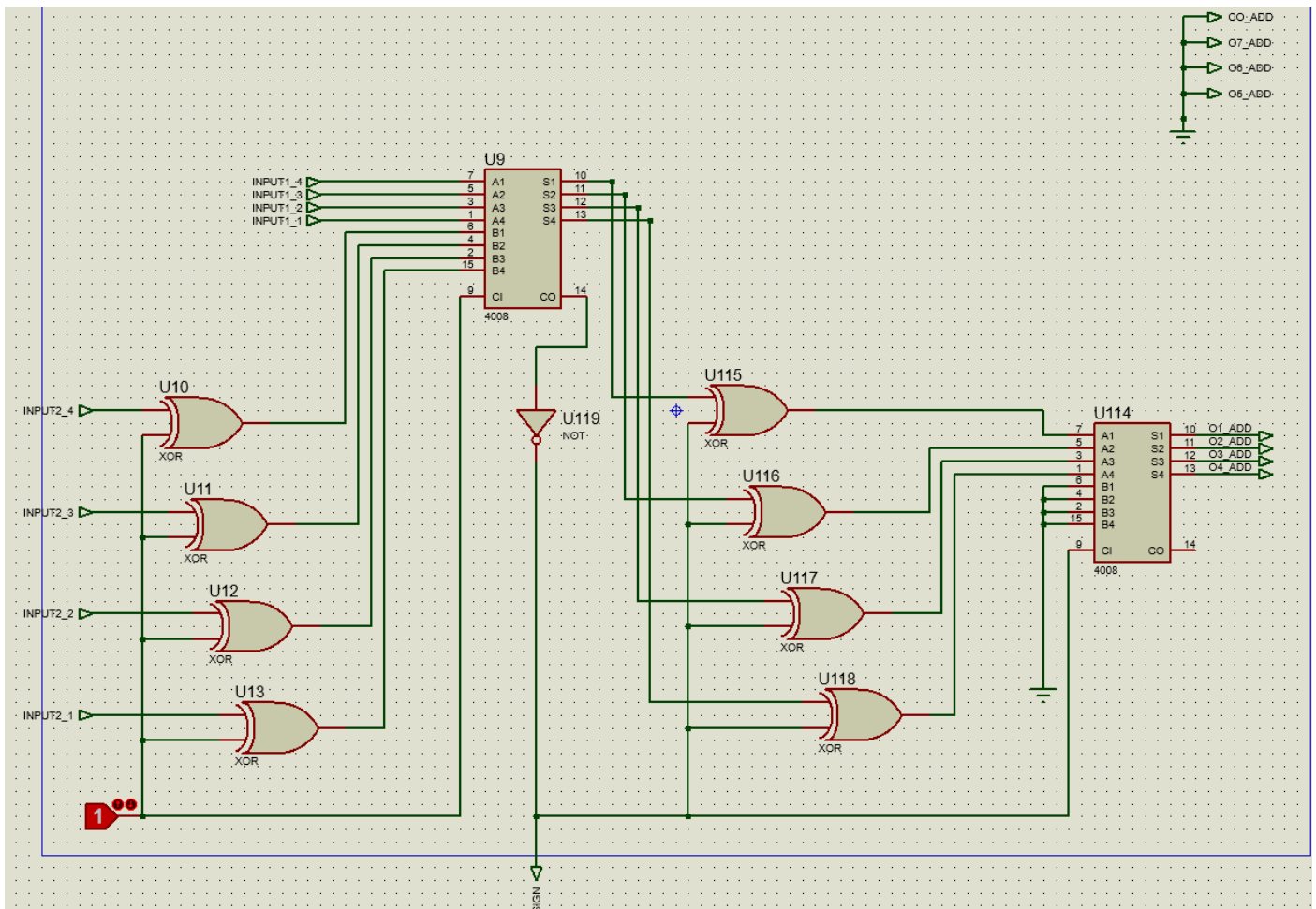
The BCD adder accepts two 4-bit BCD digits. If the binary sum exceeds 9 (1001), a correction value of 6 (0110) is added to adjust the result to valid BCD.



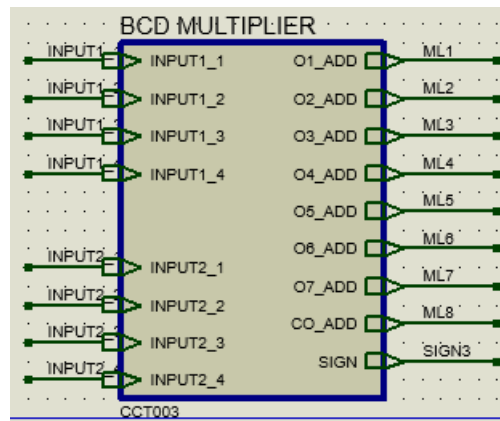
BCD Subtractor using two's complement and Correction



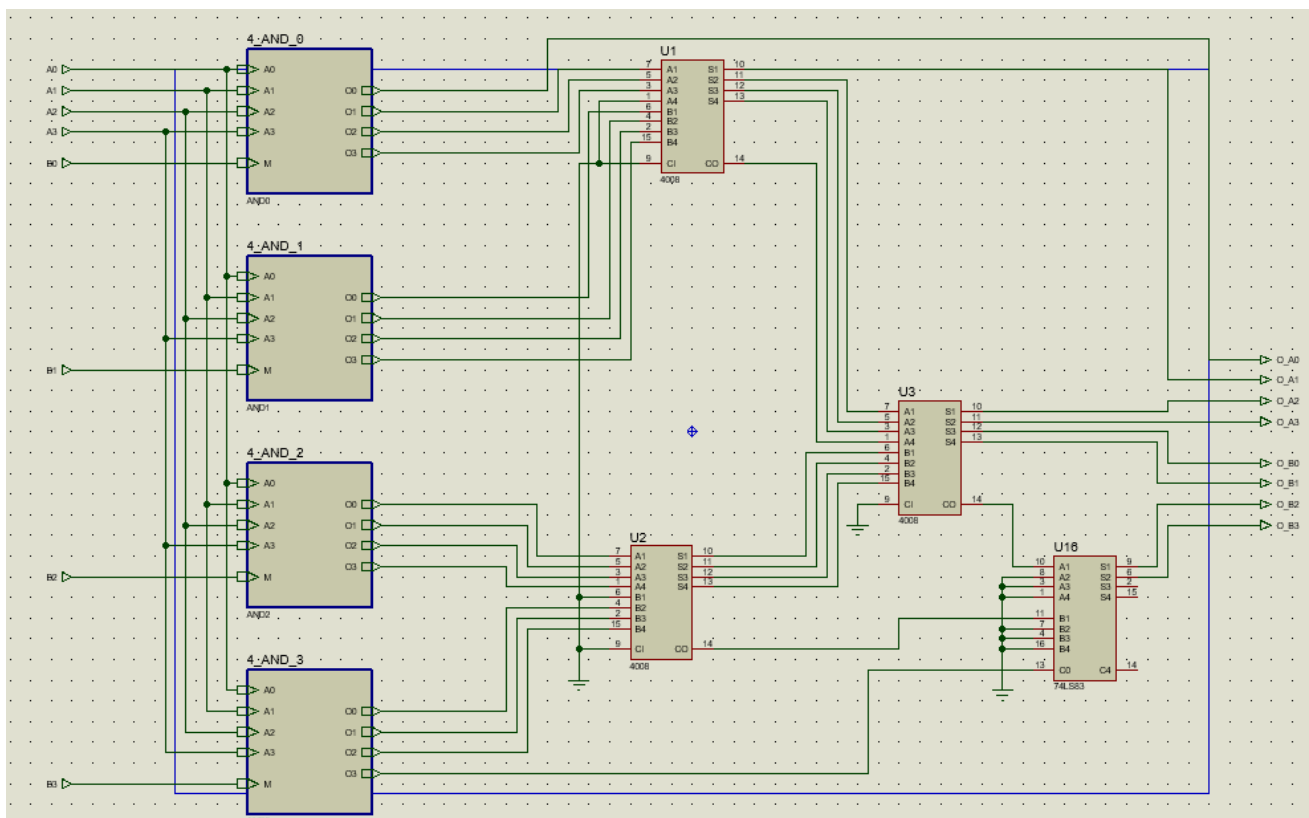
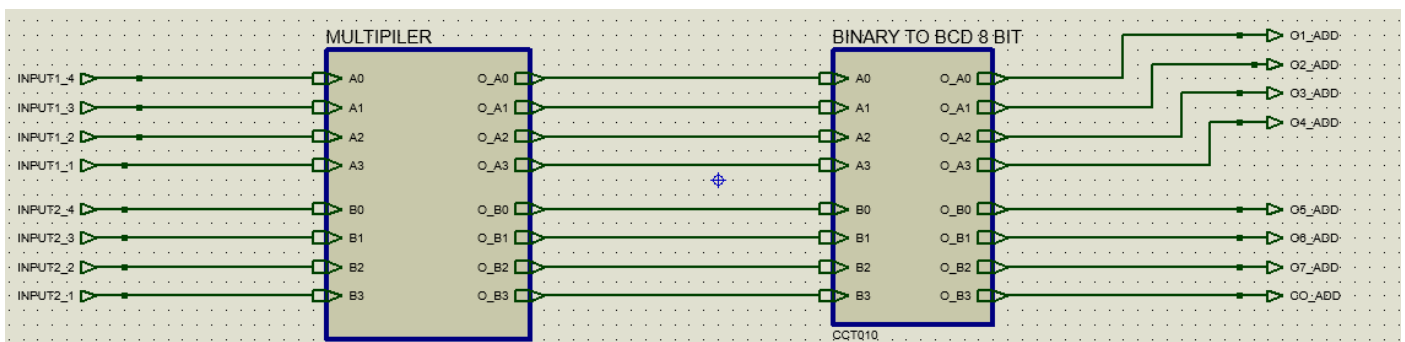
This component uses two's complement logic. The subtrahend is complemented and added to the minuend. Correction logic ensures BCD validity. Underflow (negative value) conditions are handled to maintain accuracy.



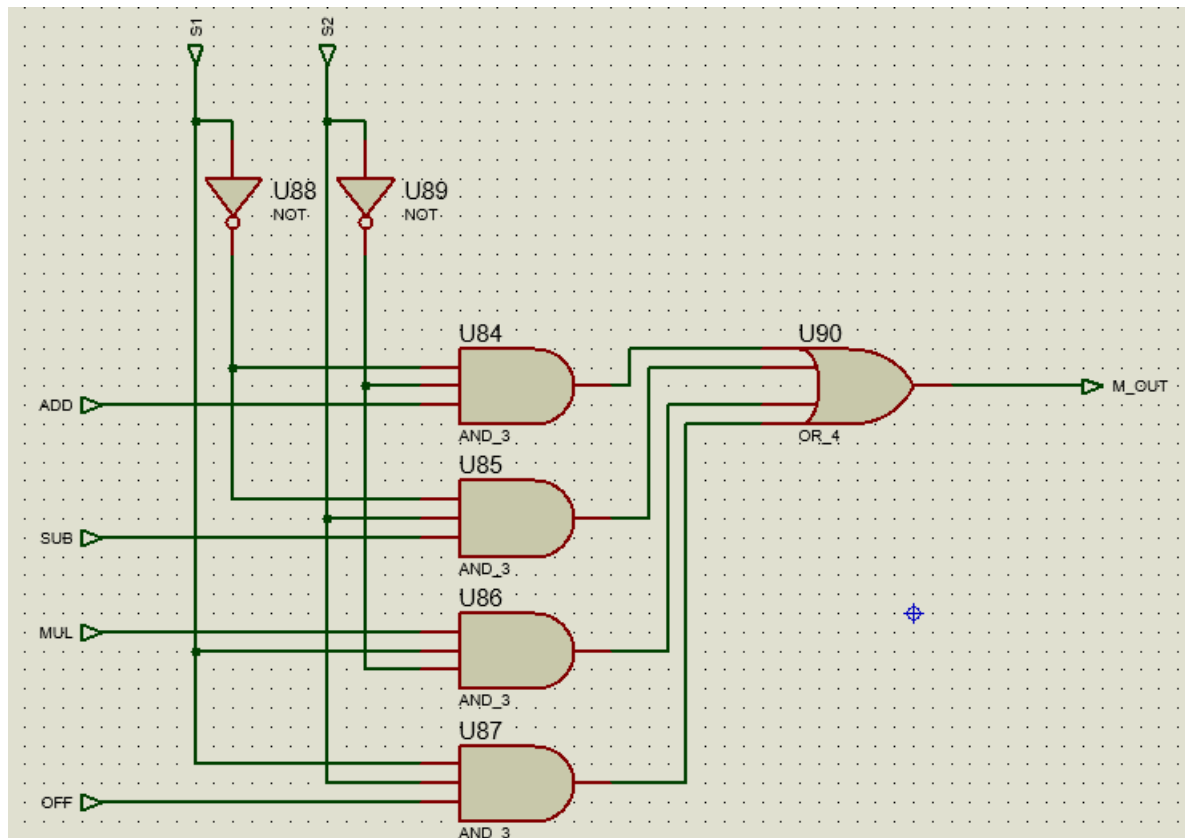
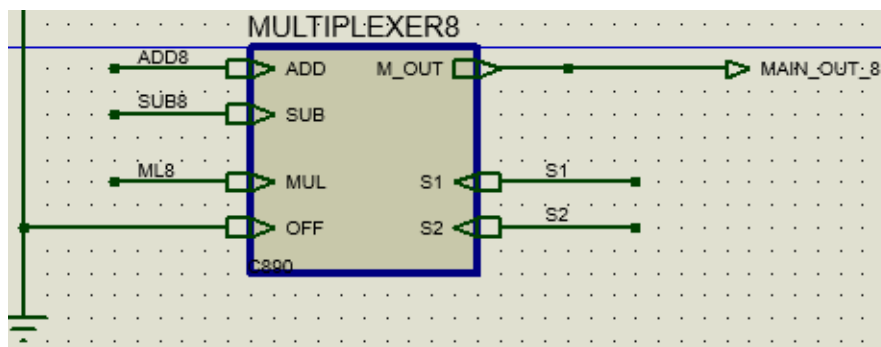
BCD Multiplier with Binary to BCD Converter



The multiplier is designed using AND gates and 4 bit adders, optimized for 4-bit BCD input. The product is transferred into the Binary to BCD Converter to ensure that final results remain in BCD format.



Multiplexer for operation control



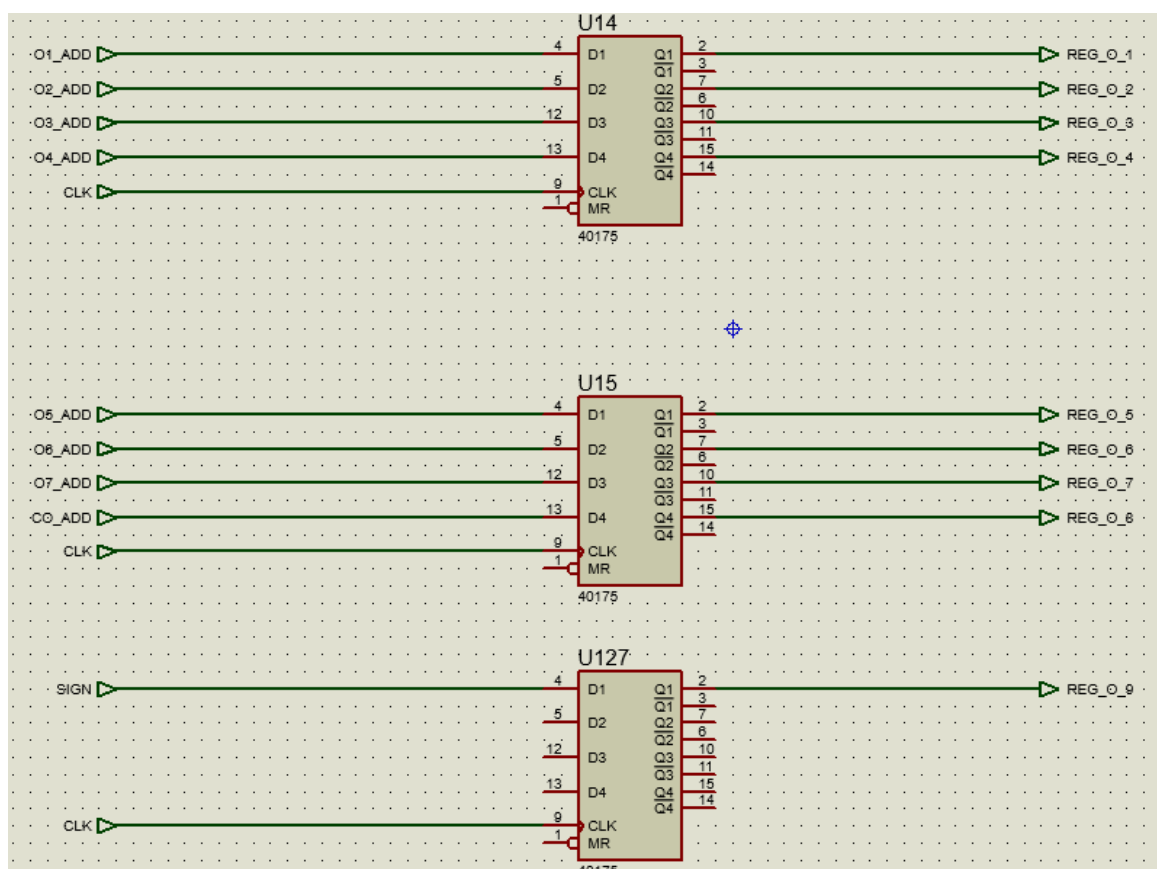
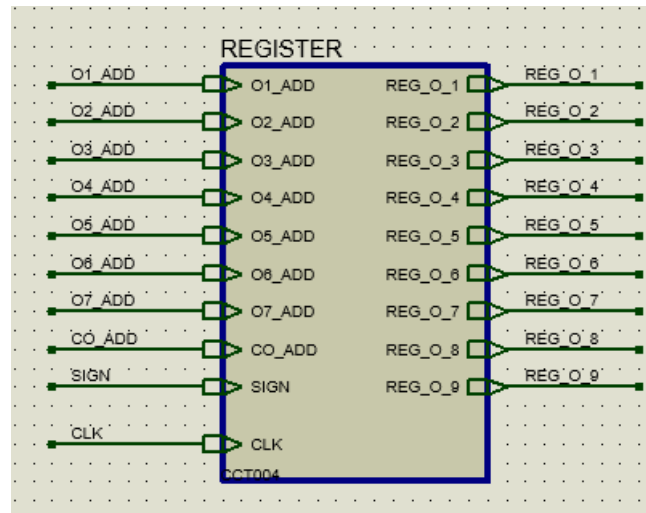
The multiplexer uses a 2-bit control signal:

- 00: Addition
- 01: Subtraction
- 10: Multiplication

The control logic interprets these signals to route inputs to the appropriate arithmetic unit and select the correct output.

Used 8 Multiplexers for Input Handling and 1 Multiplexer for Sign Handling from Subtractor.

Registers for Output Storage

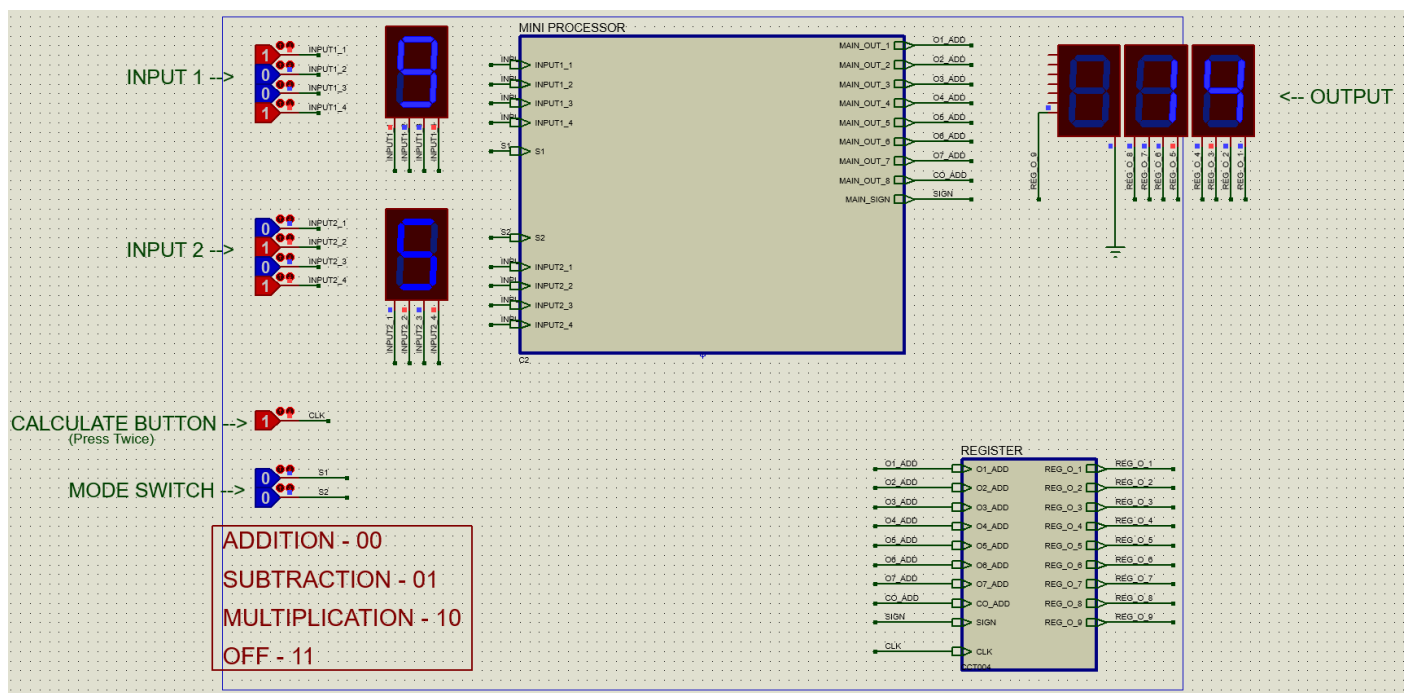
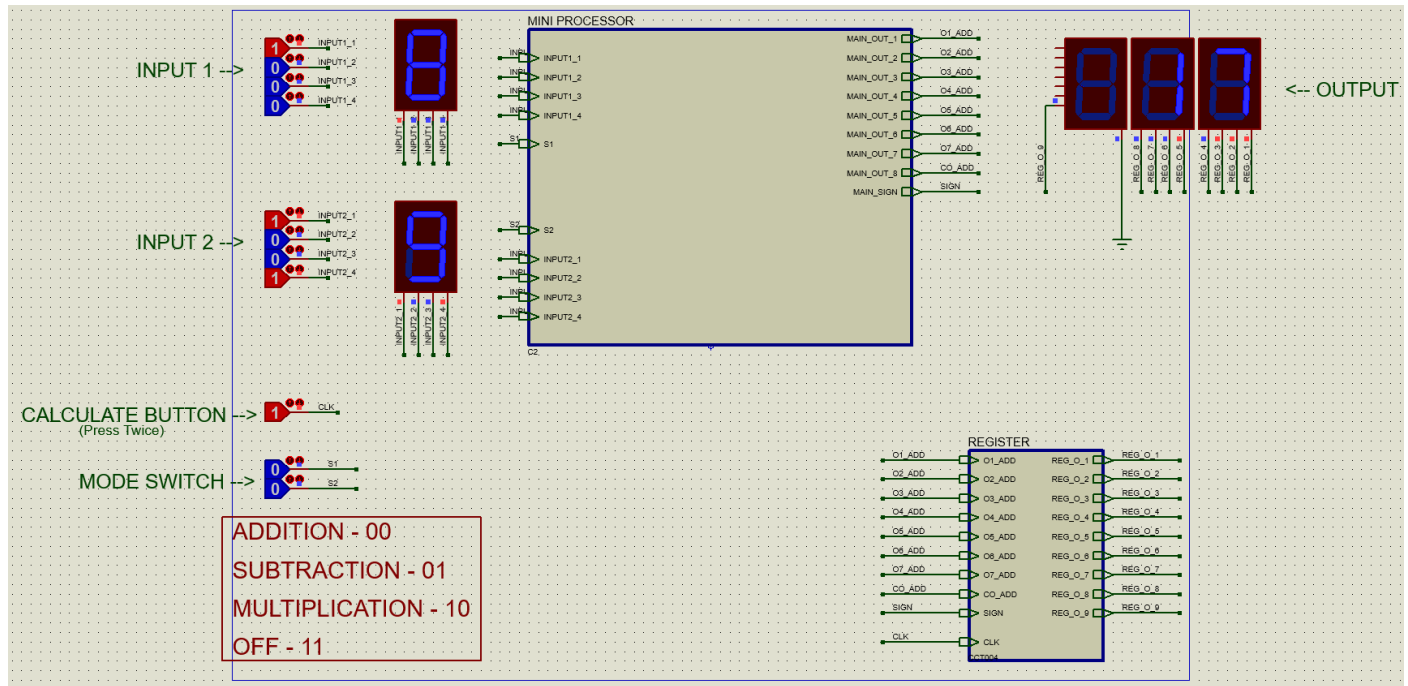


Two 4-bit registers are used to store the tens and units digits of the BCD result. They ensure stable, synchronized output and prevent glitches during transitions between operations.

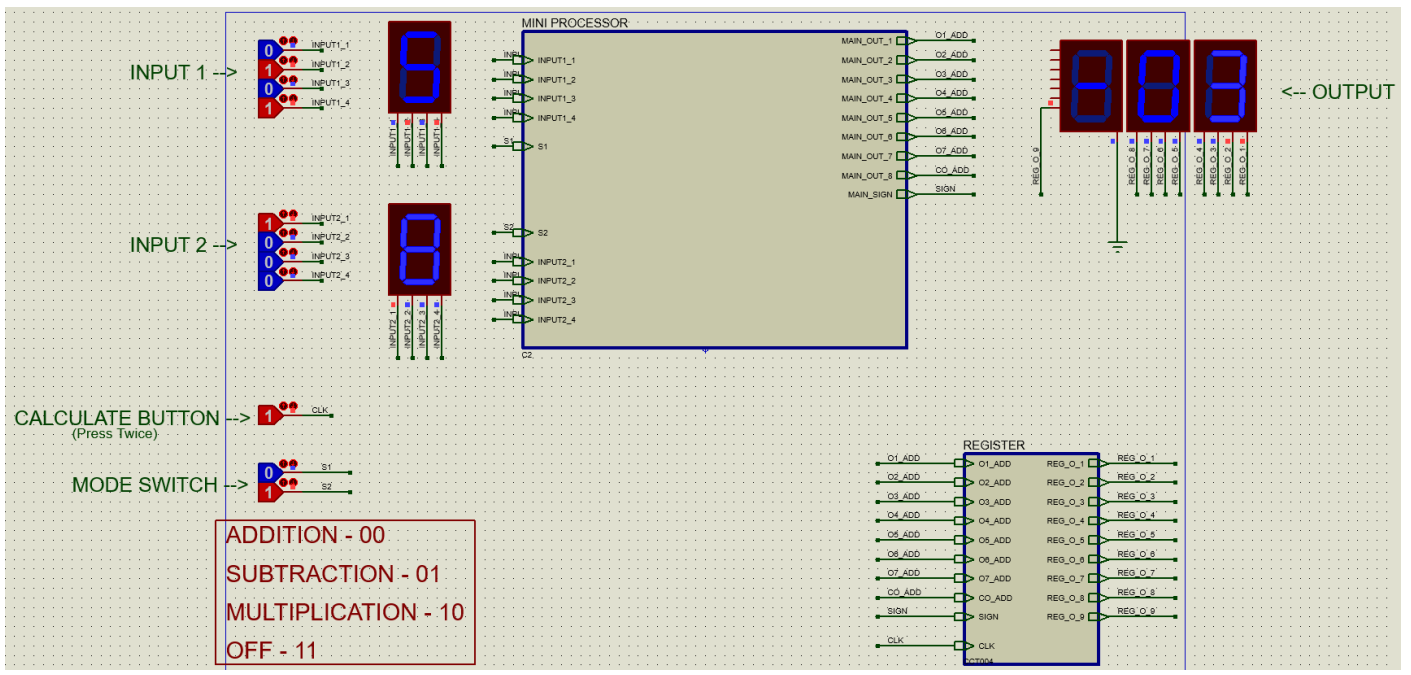
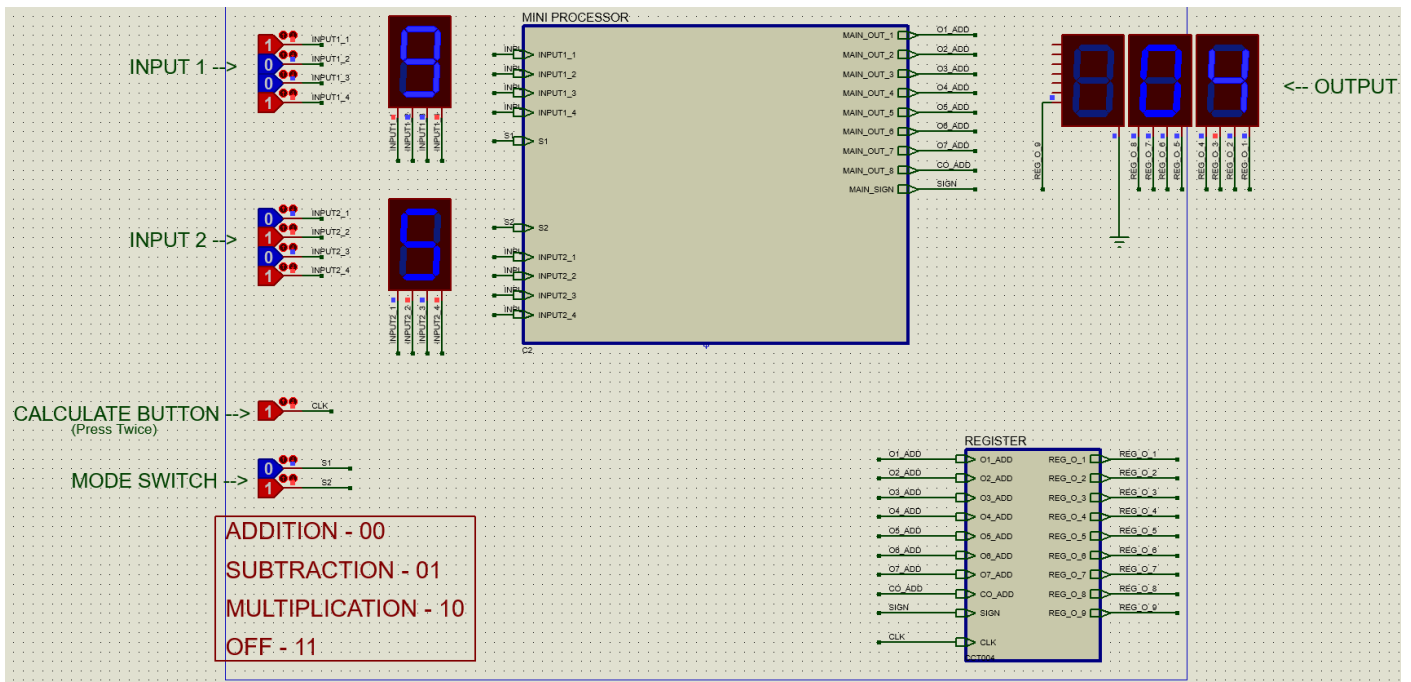
Simulation and Results

The Mini-Processor was simulated using Proteus. Each module was tested individually, followed by full-system integration.

Addition

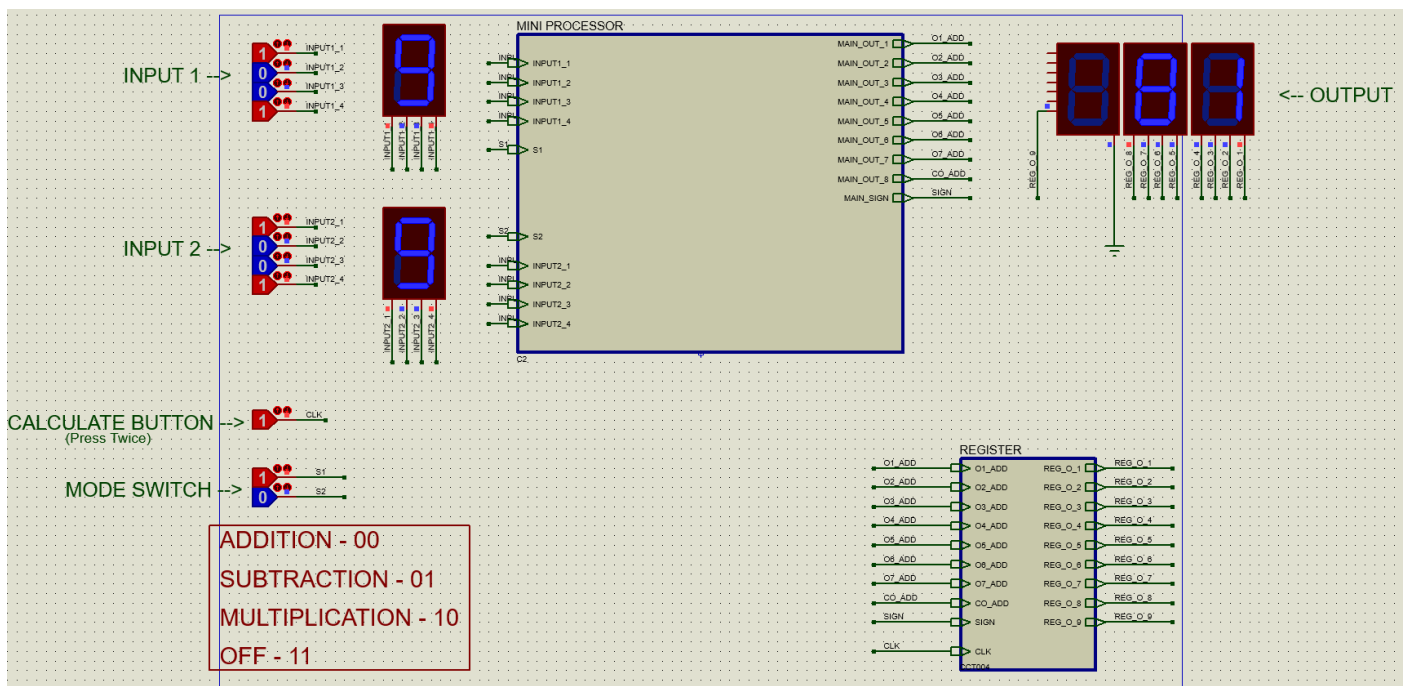
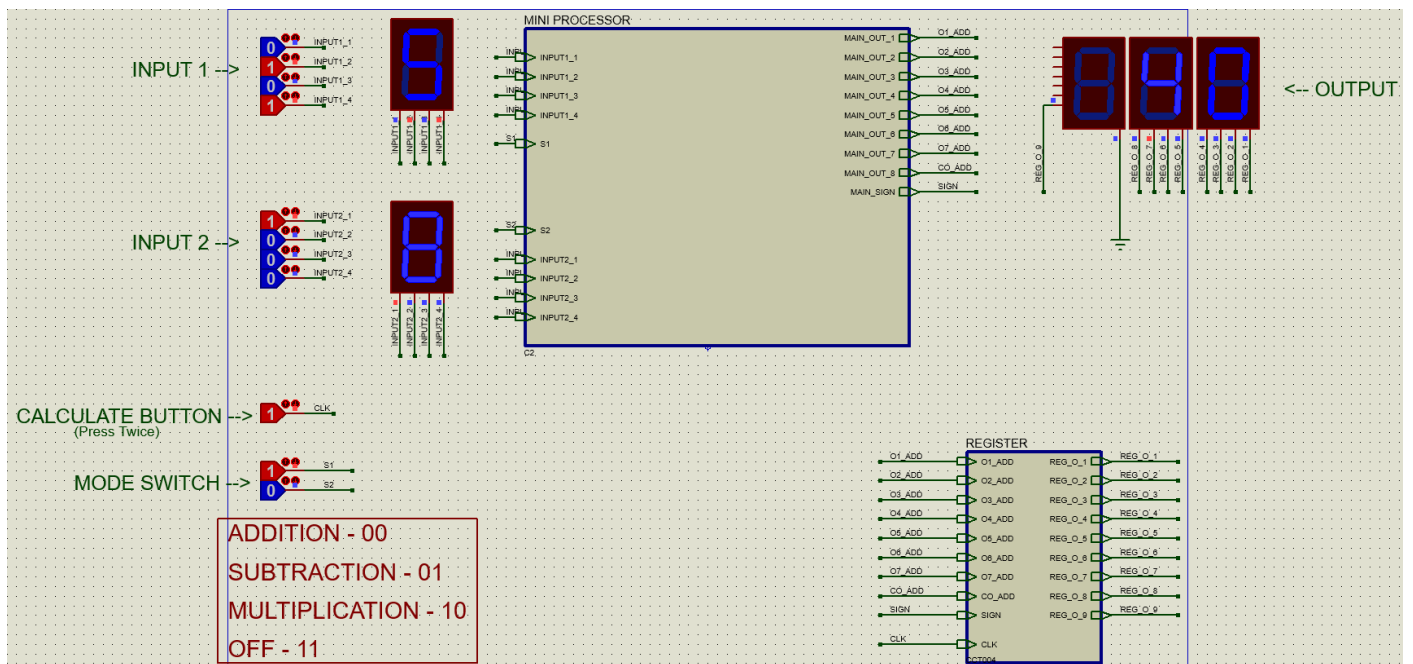


Subtraction



****When minuend is smaller than the subtrahend negative sign will appear in the front display.**

Multiplication



Challenges and Resolutions

Learning Proteus Simulation

Handling Proteus for the first time was a significant challenge. Navigating the interface, placing components, and wiring complex circuits took time to learn. With practice and reference guides, simulation skills gradually improved, allowing for accurate testing of all components.

BCD Correction Logic

Implementing correction logic in the BCD adder and subtractor was challenging. Without correction, outputs exceeded valid BCD ranges. This was resolved by checking if the binary result was greater than 9 and adding 6 to produce a valid BCD output.

Binary to BCD Conversion in Multiplication

Binary to BCD conversion was the hardest part of the project. Particularly in multiplication, the binary result had to be accurately converted into two valid 4-bit BCD digits. Achieving this required careful algorithm design and trial-and-error during simulation.

Control Logic and Multiplexer Implementation

Implementing the 2-bit control logic and correctly routing operations through the multiplexer was tricky. At first, incorrect outputs appeared due to improper input selection. This was fixed by thoroughly testing each control signal and ensuring correct mapping to the corresponding operation.

Register Implementation

Designing and integrating the registers for stable output storage posed synchronization challenges. Output values were sometimes unstable during transitions. This was solved by using clocked registers to store final results only after the computation completed, ensuring clean and accurate outputs.

Conclusion

The design and implementation of a Mini Processor capable of performing BCD addition, subtraction, and multiplication provided a comprehensive and hands-on learning experience in digital electronics. This project allowed for the practical application of theoretical concepts such as BCD representation, arithmetic circuit design, and control logic integration.

Each arithmetic unit was carefully developed with appropriate BCD correction techniques to ensure accurate results. These components were then successfully combined into a complete system using a control signal and multiplexer to manage operation selection. Registers were used to store output values, ensuring stable and glitch-free results.

Throughout the process, several challenges were encountered. The binary to BCD conversion proved to be particularly difficult due to its complexity in handling multi-digit outputs. Additionally, implementing control logic, managing proper operation routing through the multiplexer, and learning the Proteus simulation software for the first time required patience and problem-solving. These obstacles were overcome through consistent testing, logical reasoning, and gradual mastery of the tools and techniques involved.

This was the hardest assignment I have worked on so far in my academic journey. It pushed me beyond my comfort zone and significantly expanded my understanding of digital systems. I am sincerely grateful to the lecturer for assigning this task, as it challenged me to think deeply, work persistently, and grow both technically and intellectually.

The final system performed reliably across various test cases, demonstrating correct functionality and the ability to handle BCD operations efficiently. Overall, this assignment was a valuable and rewarding experience that laid a strong foundation for future projects in digital electronics and processor design.